

AURORA-12

A fully synthetic benchmark for evidence-linked poster generation

PostEx Synthetic Research Collective

Fictional Methods Laboratory · Example Research Campus · no real authors or institutions

POSTEX-SYNTH-2026-001 · fully synthetic · CC0-1.0 · no DOI

Fictional lab logo

1,240
synthetic
profiles

12
anonymous
model features

4
deterministic
cohort
simulations

Communication test

Can a realistic multi-cohort manuscript exercise evidence-linked poster generation without redistributing a real paper or implying biomedical evidence?

Evidence: Abstract and Introduction, pp. 1-2

Deterministic workflow

- 1 Fixed seed generated 1,240 profiles
- 2 480 anonymous transcript-like features
- 3 Nested elastic-net selection
- 4 12-feature score and shift testing

FOUR SYNTHETIC COHORTS

Discovery n=620 · Internal validation n=310
External simulation A n=186 · External simulation B n=124
All values are authored fixtures; no participant, specimen, institution, registry or external dataset exists.

Evidence: Abstract and Methods, pp. 1-2

Generation to validation

Synthetic benchmark workflow

All profiles, features and outcomes are authored simulation fixtures

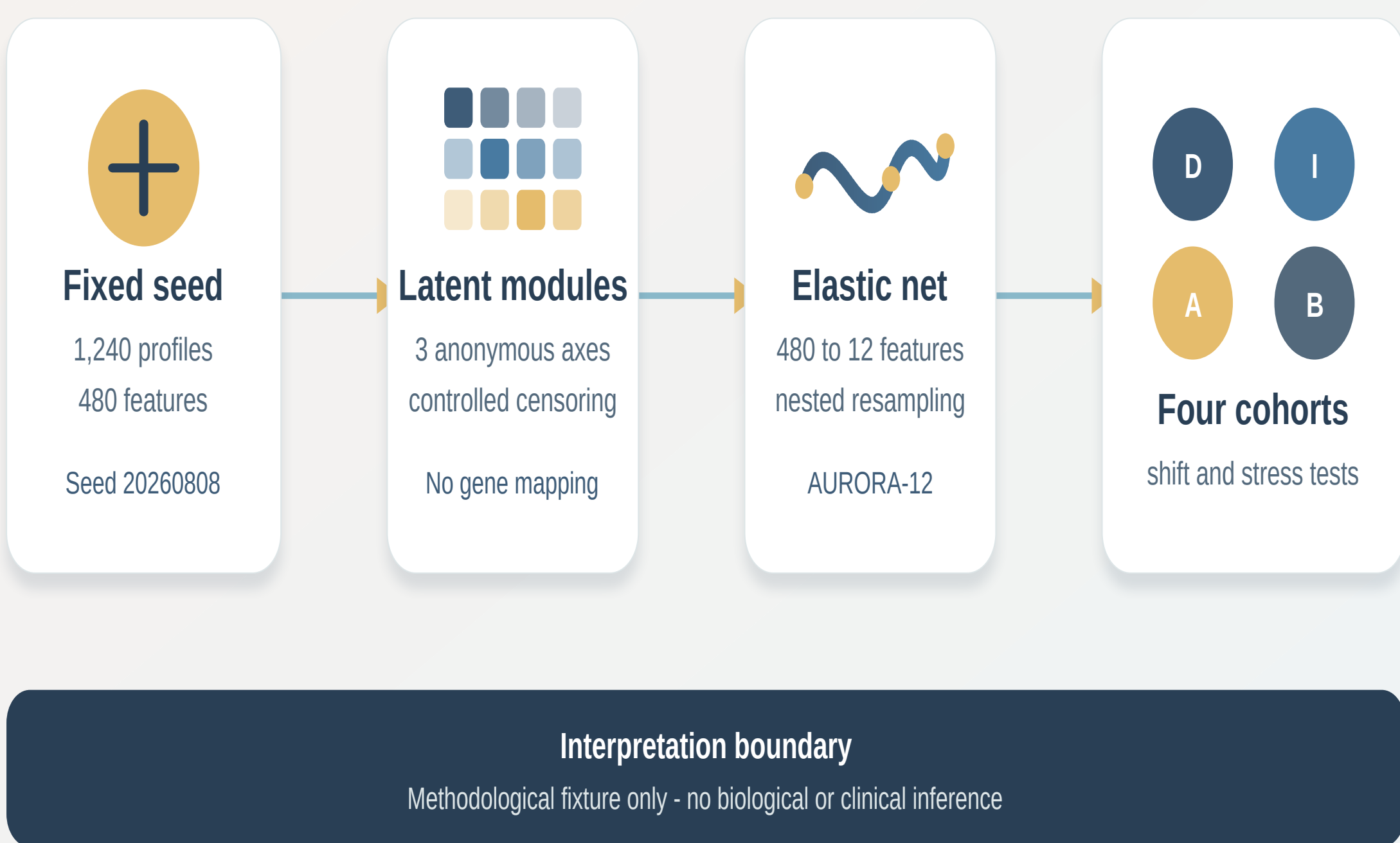


Figure 1 · Deterministic generation and validation workflow
Authored synthetic fixture · manuscript p. 2

External simulations retained C-indices of 0.75 and 0.74

Synthetic baseline: 0.66 and 0.65 · software fixture, not clinical validation

Evidence: Results, pp. 1 and 3; Figure 2

Discrimination across shifts

Authored discrimination benchmark

Concordance index - synthetic values only

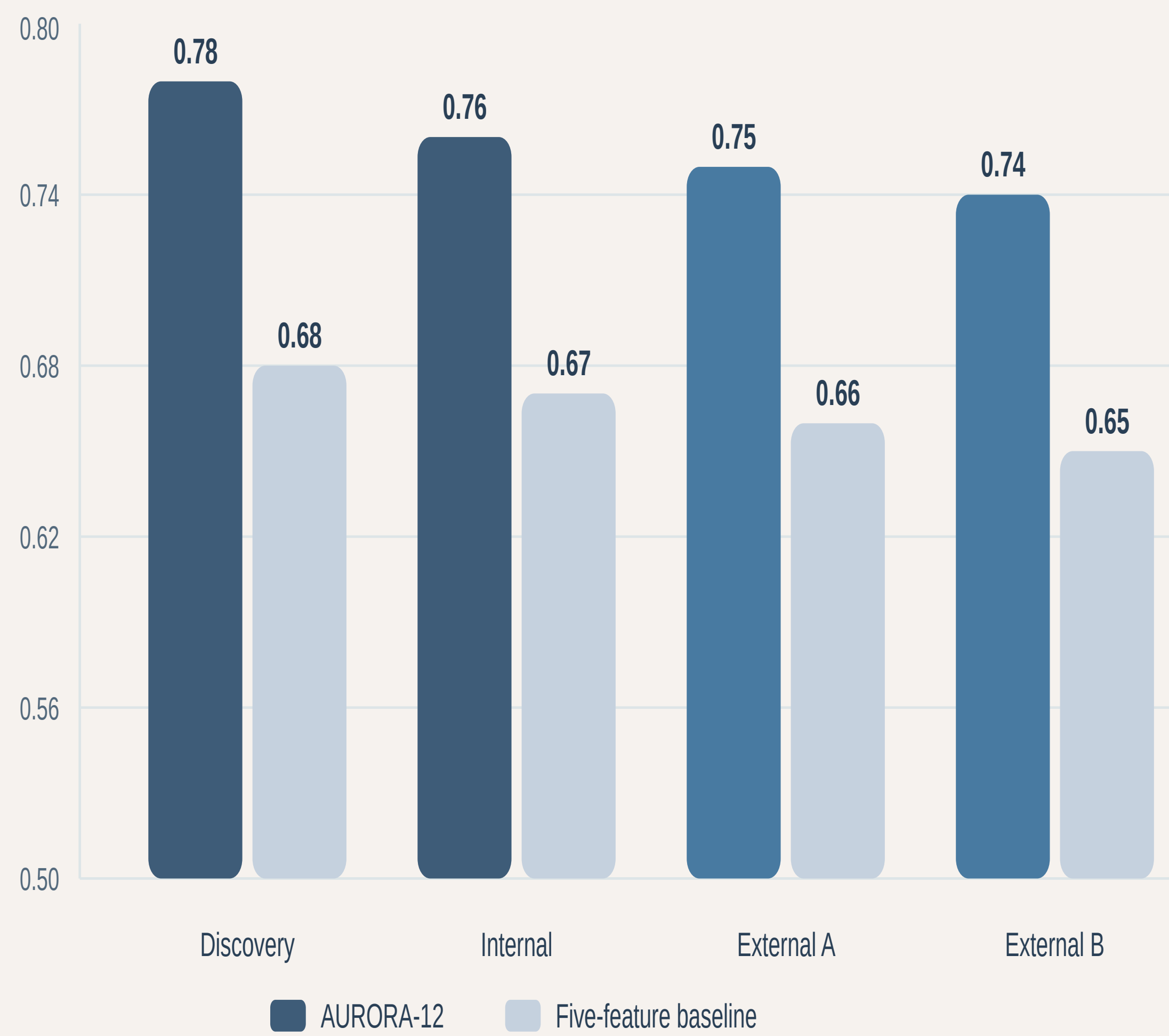


Figure 2 · Authored discrimination values across cohort shifts
Authored synthetic fixture · manuscript p. 3

Calibration and stress boundary

- External calibration slopes were 0.97 and 0.95.
- External absolute errors were 0.031 and 0.038.
- Aggregate coefficient-sign stability was 92% across 500 resamples.
- The most severe pre-specified shift reduced C-index by 0.04.

Evidence: Results, pp. 1, 3-4; Figures 3-4

AURORA-12 vs baseline C-index · Discovery 0.78 vs 0.68 · Internal 0.76 vs 0.67 · External A 0.75 vs 0.66 · External B 0.74 vs 0.65

Calibration & stability

92%

coefficient-sign stability

0.97 / 0.95

external calibration slopes

0.031 / 0.038

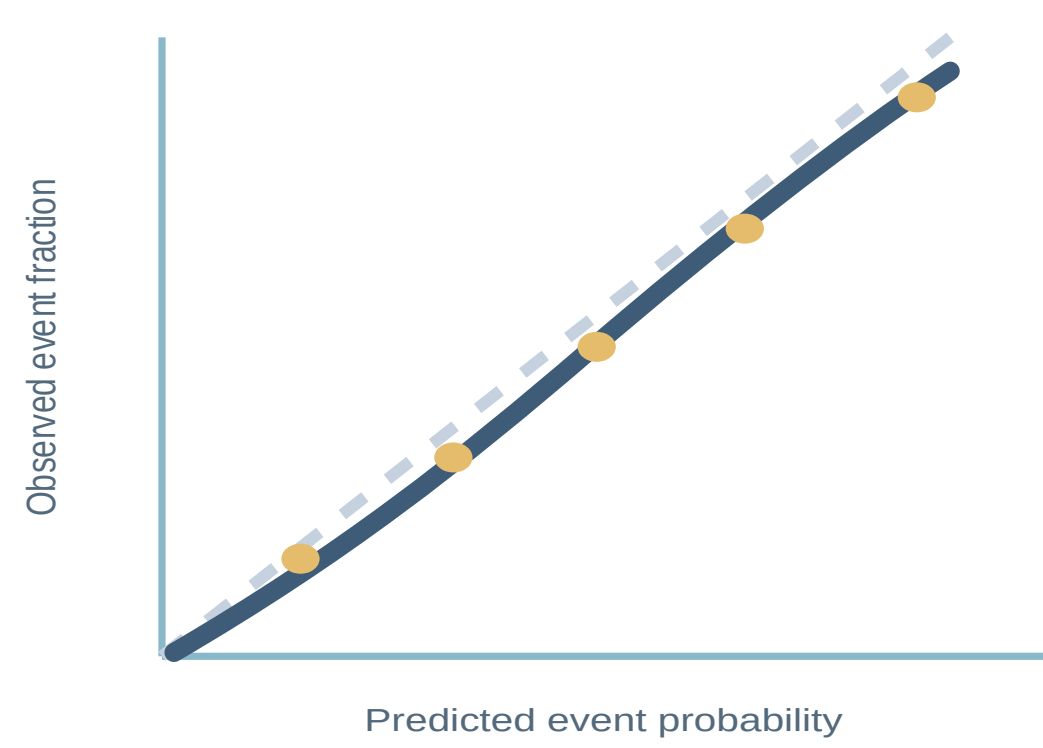
external absolute errors

Evidence: Stability and calibration, p. 4

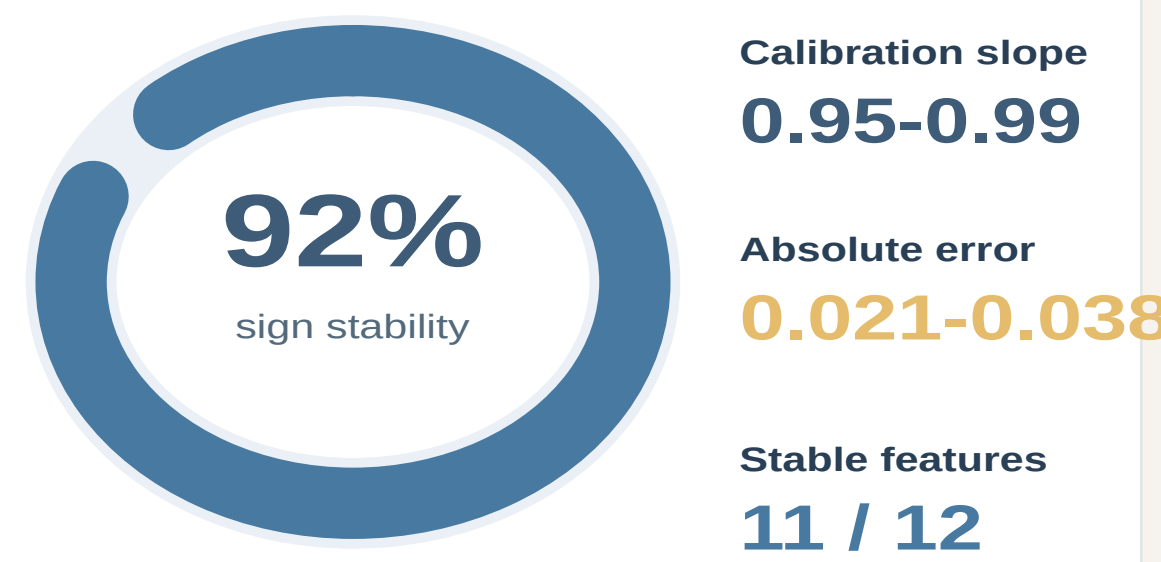
Calibration and coefficient stability

Authored evaluation values - no patient-level data

24-month calibration



500 bootstrap resamples



Synthetic stability supports fixture QA only

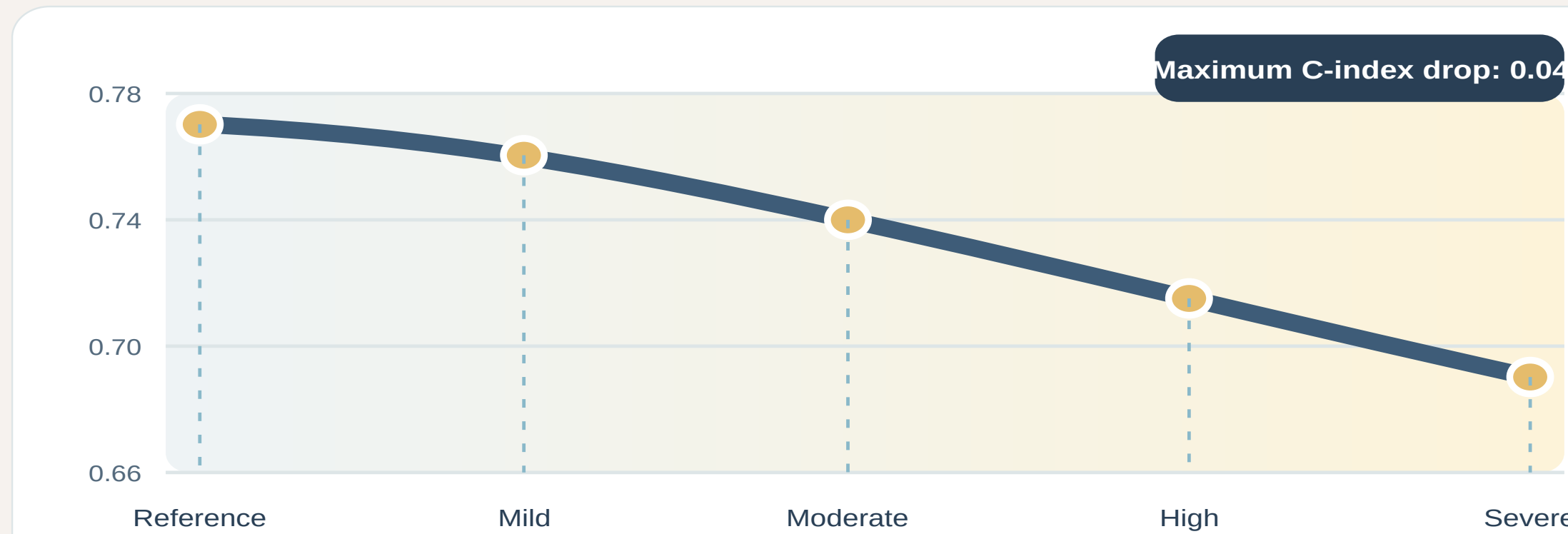
Figure 3 · Calibration and coefficient-sign stability

Authored synthetic fixture · manuscript p. 4

Pre-specified stress test

Pre-specified stress-test boundary

Increasing batch shift and latent-module imbalance



Do not interpret this authored curve as biological transportability or clinical robustness

Figure 4 · Stress-test boundary under increasing shift

Authored synthetic fixture · manuscript p. 4

Supported conclusion

AURORA-12 is a controlled, rights-safe fixture for testing evidence-linked scientific communication. It supports no inference about real biology, prognosis, treatment or clinical utility. Every number, cohort, feature, author and institution is fictional.

Evidence: Discussion and Conclusion, p. 5